

FEM Beam Analysis Tool

Nodes

Number of nodes

3 - +

Node 1 x

500.00 - +

Node 1 y

0.00 - +

Node 2 x

0.00 - +

Node 2 y

0.00 - +

Node 3 x

700.00 - +

Node 3 y

-150.00 - +

Material

Young's modulus E

20000.00 - +

Poisson's ratio ν

0.30 - +

Section

Section type

general ▾

Area

60.00 - +

Inertia

500.00

- +

Elements

Number of elements

2 - +

Element 1 start node

2 - +

Element 1 end node

1 - +

Element 1 type

Euler-Bernoulli 2-node ▾

Element 2 start node

1 - +

Element 2 end node

3 - +

Element 2 type

Euler-Bernoulli 2-node ▾

Constraints

Number of constraints

6 - +

Constraint 1 node

2 - +

Constraint 1 direction

x ▾

Constraint 1 value

0.00 - +

Constraint 2 node

2 - +

Constraint 2 direction

y ▾

Constraint 2 value

0.00 - +

Constraint 3 node

2 - +

Constraint 3 direction

rotation ▾

Constraint 3 value

0.00 - +

Constraint 4 node

3 - +

Constraint 4 direction

x ▾

Constraint 4 value

0.00 - +

Constraint 5 node

3 - +

Constraint 5 direction

y ▾

Constraint 5 value

0.00 - +

Constraint 6 node

3 - +

Constraint 6 direction

rotation ▾

Constraint 6 value

0.00 - +

Point Loads

Number of point loads

0

-

+

Distributed Loads

Number of distributed loads

1

-

+

Distributed Load 1 element

q_ini 1

q_fim 1

Direction 1

1

-

+

-0.40

-

+

-0.40

-

+

y

▼

Moment Loads

Number of moment loads

1

-

+

Moment Load 1 node

Moment 1 magnitude

1

-

+

8000.00

-

+

Run Analysis

Nodal Displacements

	Node	x	y	u	v	theta
0	1	500	0	-0.091	-0.1993	0.0687
1	2	0	0	0	0	0
2	3	700	-150	0	0	0

Download Displacements CSV

